

Credit Name: CSE 2140

Assignment Name: Chapter 3 Mastery

The following questions can help you in thinking critically about your problem-solving processes:

Understanding the Problem

How did you approach understanding the challenge?

Were there any parts of the problem you found confusing at first? If so, how did you resolve that confusion?

Exercise 4 - I first deduced that the application would be based around $E=mc^2$ formula. The user would only need to input the mass, c is the speed of light so its accounted for. I was confused on how to square the speed of light so I just multiplied it by itself.

Exercise 9 - I found having to make a calendar for this application very confusing. However I read the instructions from the textbook more closely, I could just assume that every month could be 30 days and every year is 365, so I could make a simple equation.

Planning the Solution

Did you create a plan or break the problem into smaller steps before coding?

How did you decide on the tools, data structures, or algorithms to use?

Exercise 4 - I first understood the formula, the only unknown is mass so it needs to be supplemented by the user. I decided on the structure of the code by using a input, process and output order, first asking user for mass, applying formula then displaying energy value.

Exercise 9 - I first tried to figure out how to create the equation for the calendar, I pondered whether I should create my own or use the java.time API. I decided to create my own to align with the instructions of the mastery.

Implementation

Did you write the code in small pieces or attempt the entire solution at once?

How did you test your solution along the way to make sure it was working?

Exercise 4 - I wrote the code by first by making the foundation of scanners. I used previous code created in class for the user input. I tested after each point if there was an error or program was instructing the user correctly. I also checked the calculation with another energy application to make sure it was accurate.

Exercise 9 - I first created the formula for the calendar which converted the values he user inputted for their birth date and the current date into days. Which I then calculated the difference of. This section required alot of trial and error because of the length of the equation.

Overcoming Challenges

What part of the problem was the most difficult for you?

How did you handle moments when you felt stuck or unsure of what to do next?

Exercise 4 - The most difficult part of this exercise was discovering the core of the equation, so I could understand how to calculate the amount of lightbulbs powered.

Exercise 9 - The most difficult part of the exercise was finding the placement of brackets in such a lengthy equation, I kept on receiving errors throughout the duration of coding. In order to persevere I decided to write out the equation in english to figure out where the brackets should lie.

Learning

Was there anything you learned that you think will help you with future challenges?

Exercise 4 - This exercise aided my skills in formatting mathematical equations in Java.

Exercise 9 - I believe this mastery helped in formulating my own algorithms and equations based off on just finding a purpose.